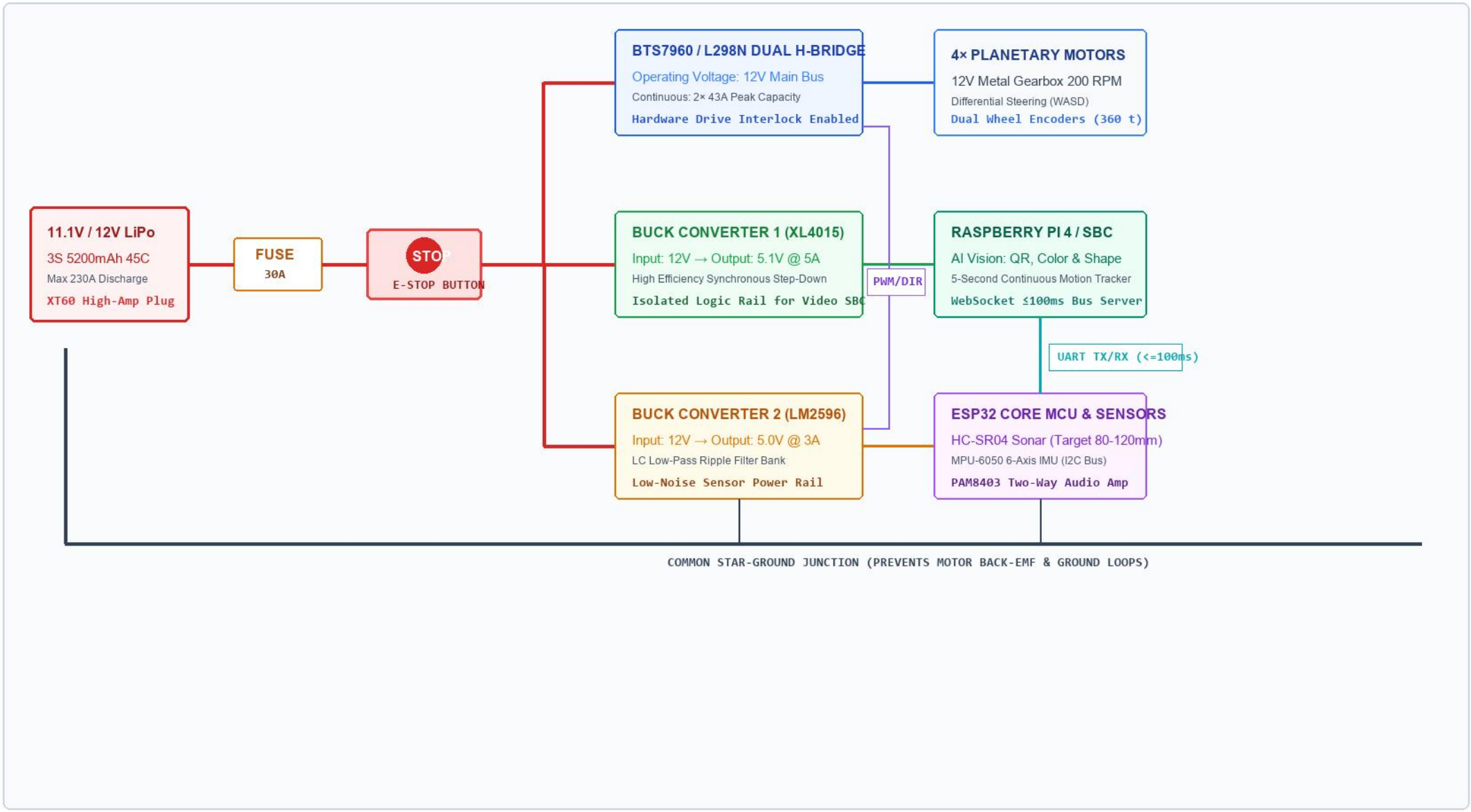
UIU RESCUE ROVER TEAM (#URRT)

ELECTRICAL SYSTEM POWER ARCHITECTURE & SCHEMATIC



1. OVERCURRENT & E-STOP (5 Marks)

30A main fuse protects against motor stall shorts. The physical e mergency stop mushroom button cuts battery power to both buck converte rs and the H-bridge drivers instantly.

2. DUAL ISOLATED STEP-DOWN RAILS

Separate step-down converters prevent motor voltage transients an d electromagnetic noise from corrupting sensitive camera vision stream s and I2C IMU bus readings.

3. STAR GROUND TOPOLOGY

All high-current motor grounds and low-power logic grounds conver ge at a single chassis star-ground junction to prevent ground loops an d false sensor readings.